

# Device Provisioning Policy

The Grand Marina Hotel — Water Monitoring System

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| Document Version | 1.0               |
| Created By       | Christine Khayat  |
| Date             | 21 September 2026 |
| Approved By      |                   |

## 1. Scope and Purpose

This policy governs the provisioning, management, and decommissioning of HYDROLOGIC devices connected to The Grand Marina Hotel's water monitoring system. It covers the full device lifecycle: adding new devices, issuing and rotating security certificates, responding to compromised devices, and removing devices from service.

### Devices Covered

| Device ID                    | Location           | Certificate    | Status |
|------------------------------|--------------------|----------------|--------|
| HYDROLOGIC-MainBuilding-001  | Mechanical Room A  | device-001.pem | Active |
| HYDROLOGIC-PoolSpa-002       | Pool Pump House    | device-002.pem | Active |
| HYDROLOGIC-Kitchen-003       | Kitchen Mechanical | device-003.pem | Active |
| [Future devices as deployed] | As Assigned        | Per Device ID  |        |

## 2. New Device Onboarding Process

### Step 1: Device Registration

Register the new HYDROLOGIC device in the hotel's device records before connecting it to the water monitoring system. Assign a unique device ID following the existing naming convention, such as **HYDROLOGIC-[Location]-[Number]**. Confirm that the device ID has not already been assigned to another device and record its physical location.

## Step 2: Certificate Generation

Generate a unique TLS client certificate for the device using `generate_client_certs.py`. The certificate must be signed by the Hydroficient Root CA and contain the device's unique identity.

The certificate generation process creates a certificate and private key for the device. The CA private key (`ca-key.pem`) must remain restricted to authorized security administrators and must never be copied to or stored on the device.

Each device must receive its own certificate and private key. Certificates must never be shared between devices.

## Step 3: Certificate Installation

*Securely transfer the device's certificate and private key to the physical HYDROLOGIC device using an authorized provisioning process. Verify that the certificate matches the intended device before installation.*

*Confirm that:*

- *The correct device certificate is being installed.*
- *The certificate belongs to the intended device ID.*
- *The private key corresponds to the certificate.*
- *The private key is kept confidential and is not shared with other devices.*

## Step 4: Connection Verification

Connect the device to the water monitoring system using mutual TLS (mTLS).

Verify that:

- The device connects to the MQTT broker on port **8883**.
- The device presents its valid client certificate.
- The broker verifies the certificate against the trusted Hydroficient Root CA.
- The device can publish its sensor data successfully.
- The device appears correctly in the monitoring system.

Test that unauthorized connection attempts are rejected. Devices without a client certificate, certificates signed by an untrusted CA, or expired certificates must not be allowed to connect.

## Step 5: Documentation

Record the following information for every provisioned device:

- Device ID
- Physical location
- Certificate filename
- Certificate fingerprint/identity
- Certificate issue and expiration dates
- Provisioning date
- Provisioning technician/security engineer
- Current device status

Update the device records whenever the certificate is rotated, the device is moved, compromised, or decommissioned.

**Estimated time for full onboarding:** 30–45 minutes/Device

## 3. Certificate Rotation Schedule

**Current certificate validity period:** 1 year

### Rotation Schedule:

- Automatic rotation every 30 days before certificate expiration
- Manual rotation triggered by 30 days warning
- Rotation before expiration with 30 days warning
- Immediate rotation if a certificate is suspected to be compromised

### Rotation Procedure:

1. Generate a new certificate and private key for the same device using `generate_client_certs.py`, signed by the trusted Hydroficient Root CA.
2. Securely install the new certificate and private key on the device and verify that they belong to the correct device.
3. Connect the device using the new certificate and confirm successful mTLS communication on port 8883.

4. After successful verification, revoke or invalidate the old certificate and update the device records with the new certificate information.
5. Confirm that the device continues sending sensor data normally after the rotation.

**Who is responsible:** The authorized security administrator/security engineering team responsible for the Grand Marina water monitoring system.

The rotation must be completed before the existing certificate expires. A device must not continue operating with an expired certificate.

## 4. Compromised Device Procedure

If a device is suspected to be compromised:

### **IMMEDIATE ACTIONS (within 15 minutes):**

1. Revoke or disable the device's client certificate so the compromised device can no longer authenticate to the MQTT broker.
2. Disconnect the affected device from the hotel network to prevent further unauthorized communication.
3. Notify the responsible security administrator and the Grand Marina water-system administrator. Record the device ID, time of discovery, and actions taken.

### **INVESTIGATION (within 24 hours):**

1. Review MQTT broker and monitoring logs for the affected device. Look for unexpected connections, unusual messages, unexpected device activity, or connections outside normal operation.
2. Determine whether unauthorized parties could have accessed sensor data or sent false readings or commands through the compromised device.

## RECOVERY:

1. Factory reset or securely reinitialize the device to remove potentially compromised credentials and configuration.
2. Generate a new certificate and private key for the device. Do not reuse credentials that may have been compromised.
3. Re-provision the device using the standard onboarding process in Section 2.
4. Verify successful mTLS connection and normal sensor data before returning the device to service.

**Emergency contact:** Grand Marina Security Administrator / Water Monitoring System Administrator

## 5. Decommissioning Process

When a device is removed from service:

### Step 1: Revoke Certificate

*Revoke or disable the device's client certificate so that the retired device can no longer authenticate to the MQTT broker. The certificate must not be reused for another device.*

### Step 2: Remove from Monitoring

*Remove the device from the authorized device records and monitoring system. Remove or disable any dashboards, alerts, subscriptions, or monitoring entries associated with the retired device.*

### Step 3: Physical Removal

*Remove the physical HYDROLOGIC device from service. Securely erase its certificate, private key, and stored configuration before the device is returned, transferred, or destroyed.*

#### Step 4: Documentation

*Update the device records to show that the device has been decommissioned. Record the decommission date, reason for removal, device ID, certificate status, physical disposition, and person responsible for the decommissioning.*

## 6. Policy Review

This policy will be reviewed:

- Annually
- After any security incident involving a HYDROLOGIC device
- When new devices are added
- When there are significant changes to the MQTT, TLS/mTLS, or certificate-management configuration
- When the certificate management process changes

**Next review date:** September 21, 2027

## Signatures

| Role                                                              | Signature        | Date              |
|-------------------------------------------------------------------|------------------|-------------------|
| Prepared by: Christine M. Khayat                                  | Christine Khayat | 21 September 2026 |
| Reviewed by: Maya Chen, Senior Security Engineer                  | Maya Chen        | 22 September 2026 |
| Approved by: Marcus Webb, General Manager, The Grand Marina Hotel | Marcus Webb      |                   |